Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

Abstract

Synthetic multi-asset equity data must reproduce each asset's return distribution and its relationship with the market. Reusing a generator fitted to full asset returns creates a problem: adding its draws to a market factor counts market variance twice. We derived a correction that centers and rescales each draw before adding the market factor, allowing reuse without fitting a second generator to regression residuals. We tested the correction on 423 non-market assets in a 424-asset United States equity and exchange-traded-fund universe, using hidden Markov generators with heavy-tailed emissions. The corrected paths retained heavy tails and recovered the calibrated market loadings with low error. On 416 complete asset histories held out from 2025, the correction improved the mean Kolmogorov–Smirnov pass rate over naive composition and brought the median ratio of synthetic to observed variance close to one. Its one-day left-tail 99% Value-at-Risk exceedance rate was similar to that of the best residual-fit method, although both exceeded the nominal rate. We also tested a jump-duration mechanism that extended visits to extreme-return states and improved volatility clustering for the broad-market exchange-traded fund with ticker SPY. The multi-asset correction remained effective with jumps enabled, but transferring the SPY jump settings improved temporal fit in training and worsened it in the 2025 holdout. The method provides a way to reuse fitted asset generators, while dependence among asset-specific residuals remains unmodeled and can lead to overstated rebalancing returns. Code, cached inputs, result summaries, and instructions for fitting the per-asset models accompany the paper.

Keywords: hidden Markov model, jump-duration mechanism, single-index model, synthetic financial data, variance correction, Value-at-Risk coverage check, volatility clustering, heavy-tailed marginals

1. Introduction

Researchers and practitioners use synthetic equity paths to test risk models, evaluate portfolio strategies, and examine market scenarios beyond those observed in a historical sample (Assefa et al., 2020; Jordon et al., 2022). A multi-asset simulator must represent both the behavior of each asset and the relationships among assets. Factor models provide a simple way to link many assets through a shared market path (Sharpe, 1963; Ross, 1976; Fama and French, 1993), but using them with existing single-asset generators requires care. The single-index model separates each asset's return into a market component and an asset-specific residual, with the market loading estimated by regression. A generator fitted to full asset returns already represents the asset's total variance. Adding its draws to the market component as though they were residuals adds market variance a second time and can also shift the mean. Fitting another generator to each asset's regression residuals avoids that problem, but requires a second fit across the asset universe. Reusing the existing generator would avoid that additional fit, provided its draws could be transformed to supply only the residual contribution. The relevant question is whether that transformation can control the mean and variance of the composed paths while retaining useful distributional and temporal behavior. Dependence among the remaining residuals is a further requirement that a per-asset adjustment alone cannot resolve.

The return features that a generator must capture are illustrated by daily excess growth rates for the broad-market exchange-traded fund with ticker SPY from 2014–2024 (Fig. 1). Most returns lie close to zero, but extreme returns occur much more often than a Gaussian model predicts (Fig. 1a,b), a widely documented feature of financial returns (Mandelbrot, 1963; Cont, 2001). Raw growth rates show little correlation across days, while absolute growth rates remain positively correlated over longer periods (Fig. 1c,d). Large and small changes therefore cluster over time, so matching the return distribution alone does not reproduce volatility dynamics. Synthetic financial data often miss these temporal properties even when the return histogram appears realistic (Stenger et al., 2024). A hidden Markov model (HMM) offers an interpretable way to represent both features: it draws returns from state-dependent distributions and uses transitions between states to represent temporal behavior (Rabiner and Juang, 1986; Rydén et al., 1998). Ordinary transitions can leave extreme states too quickly to reproduce persistent volatility, motivating an explicit duration mechanism (Bulla and Bulla, 2006). Such a mechanism can improve an individual generator, but adding a shared market path changes the distribution and temporal behavior of the composed returns. Preserving total variance therefore does not by itself establish that the single-asset improvements survive composition. Portfolio evaluation also requires dependence between assets, including relationships not explained by the market factor (Embrechts et al., 2002; Bouchey et al., 2015).

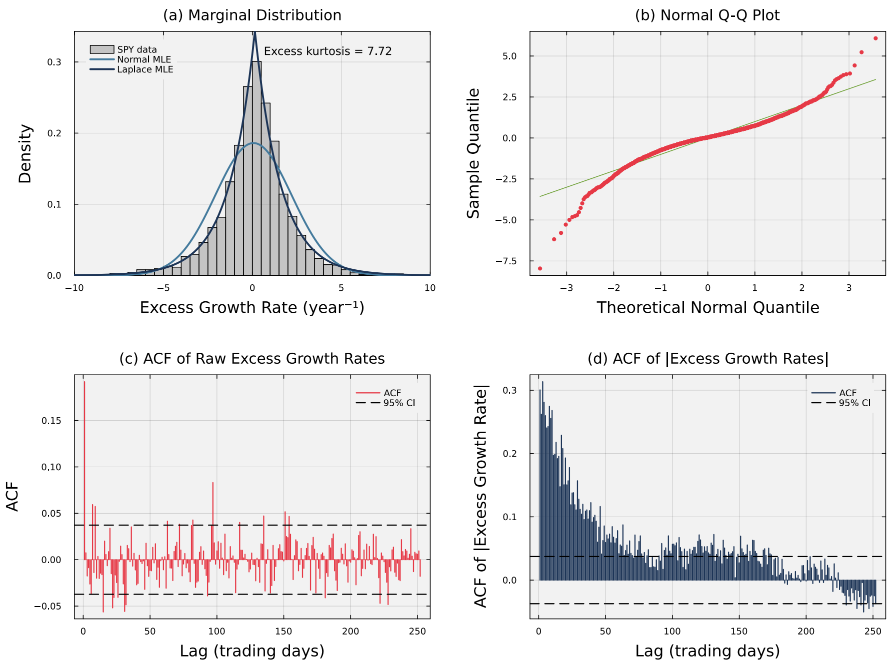

Figure 1: **Empirical stylized facts of SPY daily excess growth rates (2014–2024).** Panel (a) shows the heavy-tailed return density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) shows departure from the diagonal in a normal quantile-quantile plot at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates. Individual correlations are small, although the Ljung–Box test detects joint serial dependence. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly across the reported lags.

In this study, we developed a variance correction for combining independently fitted asset generators with a shared market path (Fig. 4; Table 3). The correction centered and rescaled full-return draws before adding the market component, avoiding a separate residual-generator fit. We used heavy-tailed hidden Markov generators and first assessed their distributional and temporal fit on SPY, including a jump-duration variant that extended visits to extreme-return states. We then evaluated composition on 423 non-market assets, initially with jumps disabled to isolate the variance correction, and compared reuse with methods fitted directly to regression residuals. With all 2014–2024 fits held fixed, the correction improved distributional fit over naive composition on 416 complete asset histories from 2025 and brought the median ratio of synthetic to observed variance close to one (Table 4). Per-asset Value-at-Risk coverage checks tested whether that improvement also produced reliable tail thresholds (Table 5). A separate comparison enabled jumps in the market alone or in both the market and asset generators to test whether the SPY temporal benefit survived composition (Table 6). The correction remained effective, but the transferred jump settings improved temporal fit in training and worsened it in the holdout. The experiments established the value and limits of generator reuse while leaving cross-asset residual dependence unresolved.

2. Related Work

Single-asset return models treat changing volatility, market regimes, and extreme moves in different ways. Autoregressive conditional heteroskedasticity (ARCH) and generalized autoregressive conditional heteroskedasticity (GARCH) models allow conditional variance to change over time (Engle, 1982; Bollerslev, 1986). Even with Gaussian innovations, the changing variance can produce an unconditional return distribution with excess kurtosis. Heavy-tailed innovations can improve the tail fit when Gaussian GARCH understates the frequency of extreme returns. Stochastic-volatility models make volatility a latent process (Stein and Stein, 1991; Heston, 1993; Kim et al., 1998). Jump-diffusion models add discontinuous price moves (Merton, 1976; Kou, 2002), but a jump component alone does not explain persistent volatility. Markov-switching ARCH and GARCH models combine discrete regimes with changing conditional variance (Hamilton and Susmel, 1994; Klaassen, 2002). Hidden-state models represent returns with a small number of market regimes. In a hidden Markov model (HMM), each latent state has a separate return distribution. Gaussian-emission HMMs can reproduce heavy tails and weak raw-return autocorrelation, but may still miss volatility clustering (Hamilton, 1989; Rydén et al., 1998). Hidden semi-Markov models allow more flexible state durations and can improve volatility persistence (Bulla and Bulla, 2006). An HMM does not create a jump process by itself. Rare extreme moves must come from the state-dependent return distributions or from a transition rule designed for such events. Deep generative models learn return distributions and temporal features without specifying a small set of market regimes or a parametric return process. Generative adversarial networks (GANs) are one class of deep generative model. Quant GANs have reproduced several financial time-series features, including volatility clustering and leverage effects (Wiese et al., 2020). Sig-Wasserstein GANs use path signatures to learn features that depend on the order of observations (Ni et al., 2021). In contrast, evaluations of GAN paths, including paths from the time-series generative adversarial network (TimeGAN), have found too little autocorrelation in absolute returns (Takahashi et al., 2019; Yoon et al., 2019; Kwon and Lee, 2024). The conflicting results show that flexible architectures do not guarantee a match to financial stylized facts. Model design, training, and evaluation all matter (Assefa et al., 2020; Stenger et al., 2024).

Multi-asset generation also requires a model of cross-asset dependence. Multivariate GARCH models estimate time-varying covariance matrices. The dynamic conditional correlation model reduces the number of parameters by separating individual volatility models from correlation dynamics (Engle, 2002; Bauwens et al., 2006). Copulas separate each asset's return distribution from the dependence model. Dynamic copulas allow cross-asset dependence to change over time, while vine copulas provide flexible high-dimensional constructions (Patton, 2006; Aas et al., 2009). Factor models reduce dimension by linking each asset to a small number of shared factors (Sharpe, 1963; Ross, 1976; Fama and French, 1993). Shared-state HMMs and multivariate GANs can also generate joint market moves (Dias et al., 2010; Rizzato et al., 2023). A standard factor model can resample historical residuals or fit a separate residual generator for every asset. This paper addresses a narrower problem. We start with single-asset generators already fitted to full returns and ask whether they can be reused in a factor model. The relevant comparison is with methods fitted directly to regression residuals: those methods remove the market component before fitting, whereas reuse adjusts an existing full-return draw during composition. Neither approach specifies cross-asset residual dependence unless that dependence is modeled separately.

3. Methods

We fitted a marginal generator for each asset and then combined the generated paths with a variance-corrected single-index model. The dataset contained 424 United States-listed equities and exchange-traded funds observed from 2014 through 2024. The single-asset analysis focused on the broad-market exchange-traded fund with ticker SPY. It used 2,767 daily volume-weighted average prices for fitting, which yielded 2,766 growth rates, and the next 250 prices from 2025, which yielded 249 holdout growth rates. The multi-asset analysis used daily closing prices. Of the 423 non-market assets in the training universe, 416 had complete 2025 histories and were included in the multi-asset holdout evaluation. We defined the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from prices $P_{i,j-1}$ and $P_{i,j}$.

$$G_{i,j} \;\equiv\; \left(\frac{1}{\Delta t}\right) \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) \;-\; r_f. \tag{1}$$

Here, $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate. We used $r_f = 0.043$ for the SPY training period and $r_f = 0.0421$ for its 2025 holdout, based on Treasury Separate Trading of Registered Interest and Principal of Securities (STRIPS) bond yields. For the multi-asset closing-price experiments, we set $r_f = 0$ in both windows, so their growth rates were annualized log returns. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.1. Per-asset hidden Markov marginal generator

For each asset, the hidden Markov model with jumps (HMM-WJ) converted the continuous return series into a sequence of N ordered return regimes. At each time step, the model first selected a regime and then drew a return from that regime's distribution (Fig. 2). We represented the model as $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ (Rabiner and Juang, 1986). The hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ identifies the regime at time t, and the observed value is the excess growth rate G_t from Eq. (1). We omit the ticker index i when discussing one asset. The transition matrix $\mathbf{T}$ sets the probabilities of moving between regimes, $\mathbf{E}$ gives the return distribution within each regime, and $\bar{\pi}$ gives the initial regime probabilities. We ordered the regimes from the most negative to the most positive returns. Their boundaries were equally spaced quantiles of a Laplace distribution fitted to the asset's in-sample returns. Specifically, $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where μ_L and b_L are the fitted location and scale. We set $Q_0 = -\infty$ and $Q_N = +\infty$ and assigned return G_t to state k when $Q_{k-1} < G_t \leq Q_k$. The Laplace distribution captured the sharp concentration of small price movements in Fig. 1a (Kotz et al., 2001; Toth and Jones, 2019). We computed its closed-form quantiles for all 424 tickers (Bilmes, 1998).

The within-state emission followed a location-scale Student-t distribution.

$$G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom. For observations assigned to state k, μ_k is the sample mean and σ_k is the sample standard deviation. A standard t_ν has variance $\nu/(\nu-2)$, so σ_k is a scale parameter rather than the emission standard deviation. At $\nu = 5$, the emission standard deviation is $\sigma_k\sqrt{5/3}$. We selected $\nu = 5$ from a sensitivity sweep. We used $\nu = \infty$ for the Normal baseline; Table 1 gives the explored range. Holding the rest of the model fixed, the Student-t emission raised simulated kurtosis toward the observed value and raised the Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates relative to the Normal emission. We also compared HMM-WJ with a hidden semi-Markov model (HSMM) baseline adapted from Bulla and Bulla (2006) (Supplementary Table S4; Online Appendix S6). The emission and HSMM comparisons separated the effects of heavy-tailed emissions and the jump-duration mechanism. We estimated the transition matrix $\mathbf{T}$ by counting observed transitions, which avoided the initialization sensitivity of iterative expectation-maximization (Bilmes, 1998). The counted matrix was concentrated near its diagonal, but individual states, including tail states, were short-lived (Supplementary Fig. S3 in Online Appendix S5). The counted transition probabilities caused the model to leave extreme-return states too quickly to reproduce the observed volatility clustering (Bulla and Bulla, 2006): a chain fitted by counting reproduces the one-step state-pair frequencies exactly, and its absolute-return autocorrelation therefore decays geometrically at a rate set by the counted matrix. Online Appendix S4 derives the fitted generator's stationary law and autocorrelation structure and quantifies both this rapid mixing and the correction added by the jump mechanism.

We added jump episodes to extend visits to extreme-return states. When no jump episode was active, a new episode began with probability ϵ. Its duration was $K \sim \text{Poisson}(\lambda)$, where λ is the mean episode length (Glasserman, 2003). Because a Poisson draw can be zero, $K = 0$ caused no forced tail step. For $K > 0$, the model temporarily replaced ordinary transitions with draws from two tail sets. The set $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ contained the most negative states, and

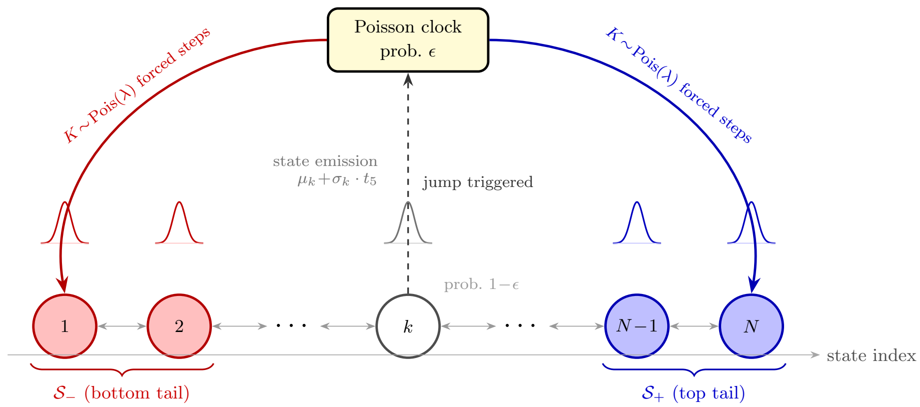

Figure 2: **Architecture of the hidden Markov model with jumps (HMM-WJ).** When no jump episode is active, the chain transitions according to the empirical transition matrix $\mathbf{T}$ with probability $1-\epsilon$ (thin bidirectional arrows) or enters a Poisson jump episode with probability ϵ (dashed upward arrow to the Poisson clock). During an episode, each of the $K \sim \text{Poisson}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise. Ordinary transitions then resume. Small bell curves above each state show the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

$\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$ contained the most positive states. At each forced step, the model selected the negative set with probability p_{neg} and the positive set otherwise. It made the tail-set choice independently at each step, so negative extreme returns occurred slightly more often than positive ones. Table 1 lists ϵ, λ, p_{neg}, and N_{tail}. Two closed-form quantities summarize the mechanism's operating point: forced steps occupy a long-run fraction of approximately $\epsilon\lambda/(1+\epsilon\lambda)$ of all steps, and a path of T steps contains at least one episode with probability approximately $1-e^{-\epsilon T}$, so ϵ sets how many paths jump while λ sets how long an episode lasts (Online Appendix S4). We obtained the initial-state distribution $\bar{\pi}$ by solving $\bar{\pi}\mathbf{T} = \bar{\pi}$ with $\sum_k \bar{\pi}_k = 1$ (Hamilton, 2018; Grinstead and Snell, 1997). Because $\mathbf{T}$ was estimated by counting, $\bar{\pi}$ equals the vector of in-sample state frequencies up to end effects, so the model's stationary growth-rate distribution is the frequency-weighted mixture of the state emission distributions. The solved $\bar{\pi}$ is exact for the ordinary transition matrix but only approximate for the jump-augmented process because it does not include the remaining jump duration. The full simulation procedure appears in Algorithm 1. Each step followed $\mathbf{T}$ or continued a jump episode and then drew a return from the selected state's emission distribution.

We chose the jump probability ϵ and mean duration λ to match the SPY absolute-return autocorrelation and kurtosis (Mandelbrot, 1963; Cont, 2001; Rydén et al., 1998). Jump-diffusion models use Poisson events to add instantaneous price jumps (Merton, 1976; Kou, 2002). In HMM-WJ, a jump episode extended a visit to the tail states. We searched a grid of (ϵ, λ) values. At each grid point,

Algorithm 1 Hidden Markov model with jumps (HMM-WJ) single-asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T; tail sets $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$.
Ensure: State sequence $\{S_t\}_{t=1}^{T}$ and growth-rate path $\{G_t\}_{t=1}^{T}$.

```
Draw S_1 ~ π̄;   draw G_1 ~ μ_{S_1} + σ_{S_1} t_5;   k ← 0      {stationary start; k: remaining forced-step counter}
for t = 2, ..., T do
    if k > 0 then
        J ← S_- w.p. p_neg, else S_+;   S_t ~ Uniform(J);   k ← k − 1      {forced tail step; sign drawn per step}
    else if Uniform(0, 1) > ε then
        S_t ~ T_{S_{t−1},·}      {standard Markov transition}
    else
        K ~ Poisson(λ)      {trigger jump; K is the episode length}
        if K = 0 then
            S_t ~ T_{S_{t−1},·}      {K = 0: no forced steps, standard transition}
        else
            J ← S_- w.p. p_neg, else S_+;   S_t ~ Uniform(J);   k ← min(K − 1, T − t)      {first forced step; K steps total, sign per step}
        end if
    end if
    Draw G_t ~ μ_{S_t} + σ_{S_t} t_5      {Eq. (2)}
end for
```

we averaged squared errors in the autocorrelation function (ACF) of absolute returns through lag L and weighted squared errors in excess kurtosis over the jump-active paths. We minimized the objective $J(\epsilon, \lambda)$.

$$J(\epsilon, \lambda) = \frac{1}{|\mathcal{R}_J|} \sum_{r \in \mathcal{R}_J} \left[\sum_{\tau=1}^{L} \left(\mathrm{ACF}_{\text{obs}}(\tau) - \mathrm{ACF}_r(\tau) \right)^2 + w_\kappa \left(\kappa_{\text{obs}} - \kappa_r \right)^2 \right], \quad (3)$$

where $\mathcal{R}_J$ is the set of jump-active simulated paths, $\mathrm{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and excess kurtosis, and $\mathrm{ACF}_r(\tau)$ and κ_r are the corresponding values for path r. We used κ for kurtosis and K for jump duration. We simulated 200 independent paths of 2,766 trading days at each grid point. A path with no jump reduces to the hidden Markov model with no jumps (HMM-NJ) and contains no information about (ϵ, λ), so the objective included only paths with at least one jump (Glasserman, 2003). Within those paths, λ controls both the height and the horizon of the slow autocorrelation component while the kurtosis penalty limits the total forced-step mass; ϵ acts mainly through the share of paths that contain an episode (Online Appendix S4). The value of w_κ set the relative weight of the kurtosis error. The search ranges, w_κ, L, and the remaining HMM-WJ and SIM hyperparameters are reported in Table 1. We fixed the selected hyperparameters before running the generator comparisons and the holdout evaluation.

3.2. Variance-corrected multi-asset composition

The single-asset model generated each ticker separately and did not specify cross-asset dependence. For the multi-asset model, we added a shared market path as a common factor. We used the single-index model (SIM) of Sharpe (1963).

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T. \tag{4}$$

Here α_i is the intercept for asset i, g_m is the market growth-rate path, β_i is the asset's market loading, and ε_i is its asset-specific residual path. We use ε_i for the residual path and ϵ for the jump probability in the single-asset model. We estimated the intercept and market loading (α_i, β_i) from real data by ordinary least squares (OLS). The choice of ε_i determined the composition method. A standard SIM draws independent Gaussian residuals with the variance estimated by OLS. These Gaussian residuals preserve the fitted market relationship but do not retain the heavy tails and volatility regimes produced by the single-asset generator. We called the Gaussian-residual method *Gaussian SIM.*

Naive full-path composition. In the naive construction, a full synthetic return path $\tilde{g}_i$ from the single-asset generator served as the residual.

$$g_i = \alpha_i + \beta_i\, g_m + \tilde{g}_i. \tag{5}$$

The fitted intercept and market term already set the conditional mean of asset i, while the full-return draw generally has $\mathbb{E}[\tilde{g}_i] \neq 0$. Equation (5) therefore has conditional mean $\alpha_i + \mathbb{E}[\tilde{g}_i] + \beta_i g_m$, not $\alpha_i + \beta_i g_m$. We centered every full-return path before composition to remove the duplicate mean. Centering subtracted the realized path mean.

$$\bar{\tilde{g}}_i \equiv \frac{1}{T} \sum_{t=1}^{T} \tilde{g}_i(t), \qquad \tilde{g}_i^{\mathrm{c}}(t) \equiv \tilde{g}_i(t) - \bar{\tilde{g}}_i. \tag{6}$$

Therefore, $T^{-1} \sum_t \tilde{g}_i^{\mathrm{c}}(t) = 0$ exactly. Subtracting a constant leaves both the sample variance and the sample covariance with g_m unchanged. We used the centered construction $g_i = \alpha_i + \beta_i g_m + \tilde{g}_i^{\mathrm{c}}$ as the *naive* benchmark.

Centering corrected the mean but not the variance. The path variance $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}(\tilde{g}_i) = \mathrm{Var}(\tilde{g}_i^{\mathrm{c}})$ is the *generator variance.* It already represents the asset's total variance. Adding the market term therefore adds variance a second time. We sampled the residual and market paths independently. Their covariance is zero in population and close to zero on a long realized path. We approximated the naive composed variance accordingly.

$$\mathrm{Var}(g_i) \;\approx\; \beta_i^2 \sigma_m^2 + \sigma^2_{\mathrm{gen},i}, \qquad \mathrm{Cov}(\tilde{g}_i^{\mathrm{c}}, g_m) \approx 0, \tag{7}$$

where $\sigma_m^2 = \mathrm{Var}(g_m)$ is the variance of the market path. The excess is $\beta_i^2 \sigma_m^2$, so the error is largest for assets with large market loadings.

Variance correction. We set $\sigma^2_{\text{gen},i}$ as the target total variance for asset i. The market term contributed $\beta_i^2\sigma_m^2$, so the residual term could contribute only the remaining variance. We defined s_i as the nonnegative asset-specific residual scale, set $\varepsilon_i = s_i\tilde{g}_i^{\text{c}}$, and required $\text{Var}(\beta_i g_m + s_i\tilde{g}_i^{\text{c}}) = \sigma^2_{\text{gen},i}$. Under the zero-covariance assumption, solving for s_i gives the correction.

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\text{gen},i}}} \tag{8}$$

The *market loading ratio* ρ_i is the market variance contribution divided by the target variance.

$$\rho_i \;\equiv\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\text{gen},i}}, \tag{9}$$

Under the zero-covariance assumption, ρ_i is the fraction of the target variance contributed by the market, and $s_i^2 = 1 - \rho_i$. At $\rho_i = 1$, the unconstrained rule leaves no asset-specific variance; for $\rho_i > 1$, it has no real-valued solution. Values of ρ_i at or above one can arise when β_i is large, the generator variance is low, or the synthetic market is more volatile than the calibration market. We required the asset-specific draw to contribute at least a fraction f of the target variance. We set $\rho_i^{\max} = 1 - f$ and reduced the effective loading β_i^{eff} when $\rho_i > \rho_i^{\max}$. We applied the piecewise correction in Eq. (10).

$$\beta_i^{\text{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \le \rho_i^{\max} \\ \text{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma^2_{\text{gen},i}}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \le \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{10}$$

The correction retained the sign of the calibrated loading β_i. We used $f = 0.10$, so the asset-specific draw contributed at least 10% of the target variance for non-tracker assets (Table 1). Online Appendix S9.1 examines the substitution of a full-return draw for the residual: it derives what the construction preserves exactly and how the composed shape dilutes as ρ_i grows.

High-R^2 index trackers. The non-tracker branch targeted total generator variance. For an index-tracking fund, we instead targeted its fitted relationship with the market. The coefficient of determination R^2 measures the fraction of the asset's variance explained by the market. For example, SPY regressed on itself has $\beta = 1$, $R^2 = 1$, and no residual variance. QQQ and SPYG also had high R^2 against SPY. The non-tracker branch can change the fitted R^2 even when it preserves total variance. We therefore used a separate tracker branch when $R^2_{i,\text{real}}$ was at least R^2_{preserve}. We set $R^2_{\text{preserve}} = 0.80$, which separated the index trackers from individual stocks in our universe (Supplementary Fig. S5). For the tracker branch, we computed the residual variance required to recover the calibrated R^2 under the zero-covariance assumption.

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1 - R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{11}$$

We rescaled the generator draw to the target residual variance.

$$\varepsilon_i \;=\; \left(\sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\right)\tilde{g}_i^{\text{c}}. \tag{12}$$

When $R^2_{i,\text{real}} = 1$, the target residual variance is zero. SPY regressed on itself therefore has no residual term. A scale factor greater than one increases the variance of the generator draw and can weaken the original marginal fit.

Complete composition rule. The tracker branch targeted the calibrated R^2. The non-tracker branch targeted the generator variance and reduced β_i only when the market term would leave less than the required asset-specific variance. After selecting a branch, we added the scaled asset draw to the market term. The full procedure appears in Algorithm 2. If used, a residual-copula rank reorder was applied jointly after scaling and before composition. The reorder preserved the empirical variance of each residual path. Centering kept the path mean controlled by the intercept and market term. Under the zero-covariance assumption, OLS estimates from longer composed paths converge to the effective market loading and its intercept. The variance and loading properties are asymptotic. They do not specify exact finite-sample values or the complete return distribution. Online Appendix S9 gives the algebra. The construction links assets through same-day market returns. A simulated market episode can therefore affect several assets together, but the rule does not separately fit lead–lag effects or a joint volatility process.

Algorithm 2 Hybrid single-index model (SIM) composition with variance correction.

Require: Nonconstant $g_m, \tilde{g}_i \in \mathbb{R}^T$, $i = 1, \dots, N$; each $\tilde{g}_i$ is independent of g_m.
Require: Parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1]$.
Ensure: Composed paths $g_i \in \mathbb{R}^T$ and branch flags b_i.

$\sigma_m^2 \leftarrow \text{Var}(g_m)$.
for each asset $i = 1, \dots, N$ **do**
  $\sigma^2_{\text{gen},i} \leftarrow \text{Var}(\tilde{g}_i)$; $\tilde{g}_i^{\text{c}} \leftarrow \tilde{g}_i - \overline{\tilde{g}}_i$.
  if $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
    $v_i \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}})/R^2_{i,\text{real}}$.
    $s_i \leftarrow \sqrt{v_i/\sigma^2_{\text{gen},i}}$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; $b_i \leftarrow$ R2-PRESERVE.
  else
    $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$
    if $\rho_i \leq 1 - f$ **then**
      $s_i \leftarrow \sqrt{1 - \rho_i}$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; $b_i \leftarrow$ VARIANCE.
    else
      $s_i \leftarrow \sqrt{f}$; $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i)\sqrt{(1-f)\sigma^2_{\text{gen},i}/\sigma_m^2}$.
      $b_i \leftarrow$ CLIPPED.
    end if
  end if
  $\varepsilon_i \leftarrow s_i \tilde{g}_i^{\text{c}}$.
end for
Optional: jointly rank-reorder $\{\varepsilon_i\}_{i=1}^N$.
for each asset $i = 1, \dots, N$ **do**
  $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}} g_m + \varepsilon_i$ {Eq. (4)}
end for

Table 1: **Hidden Markov model with jumps (HMM-WJ) and single-index model (SIM) composition hyperparameters.** Values used for the main SPY analysis, explored ranges, and locations of sensitivity checks. For the primary six-method SIM comparison, we set $\epsilon = 0$ instead of the SPY value. The separate jump-enabled comparison transferred the listed SPY settings to enabled models. The cross-asset validation used per-ticker (ϵ^*, λ^*) values for NVDA, JNJ, and JPM (Supplementary Table S10).

Symbol	Description	SPY value	Explored range
State partition and emission			
N	Laplace-quantile states	100	$\{30, \dots, 350\}$, Online Appendix S7
ν	Student-t emission degrees of freedom	5	$\{3, \dots, 30, \infty\}$
Jump-duration override			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	90	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	autocorrelation lag window, Eq. (3)	25 days	fixed
SIM composition			
f	idiosyncratic variance floor	0.10	$\{0.05, \dots, 0.30\}$
R^2_{preserve}	branch-selection threshold	0.80	$\{0.70, \dots, 0.90\}$

Evaluation protocol. We evaluated the method on 424 United States equities and exchange-traded funds from Polygon.io. Each ticker had 2,767 daily closing prices from January 2014 through December 2024, which yielded 2,766 growth rates. This complete-case selection conditioned the analysis on assets that survived and retained data coverage throughout the training window. We used SPY as the market and composed the other 423 assets. For each asset, we fitted the state and emission model used in the SPY analysis (Table 1; $r_f = 0$). In the primary six-method comparison, we set the jump probability to zero to measure composition without per-ticker jump calibration. We regressed each real asset on SPY to estimate α_i, β_i, $R^2_{i,\text{real}}$, and residual variance. Across the sample, β_i ranged from 0.03 to 1.79, and $R^2_{i,\text{real}}$ ranged from 0.001 to 0.933 (Supplementary Figs. S4 and S5). We compared six ways to construct the asset-specific term (Table 3). The naive method used the centered full-return draw without variance correction. The Gaussian SIM method used independent Gaussian residuals with variance estimated by OLS. The corrected method, labeled *hybrid* in the tables and figures, used Algorithm 2. The JumpHMM-on-residuals method fitted a new JumpHMM to the OLS residual series after converting the residuals to a synthetic price path by cumulative compounding. The block-bootstrap method sampled the OLS residual series with the Politis–Romano stationary bootstrap (Politis and Romano, 1994) and a mean block length of $\sqrt{T} \approx 50$. The GARCH(1,1)-t method fitted a conditional-variance model to each residual series (Bollerslev, 1986). We excluded GARCH fits that violated the stationarity condition. For each ticker, the naive and corrected methods received the same JumpHMM draw, which isolated the effect of the variance correction. The other four methods generated their own residual draws. JumpHMM names identify the generator implementation; its jump mechanism was disabled in both full-return and residual fits for this six-method comparison. We did not use the optional cross-sectional rank reorder because the experiment tested each ticker's marginal construction separately.

We evaluated each composed series on factor recovery, distributional fit, and variance preservation. For factor recovery, a new OLS regression produced the absolute errors $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$; lower values indicate closer recovery. For distributional fit, we used Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates, Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index. Higher KS and AD pass rates and lower Wasserstein distance indicate a closer match; kurtosis and the Hill index describe tail behavior. For variance preservation, we used the composed-to-generator variance ratio. A ratio of one indicates equal variances. Online Appendix S3 gives the metric definitions. We ran 100 replications per ticker with seed 1234, producing 42,300 series per method. Pass rates used the significance level $\alpha = 0.05$. For the six-method holdout evaluation, we froze all marginal fits, residual models, and SIM parameters estimated from 2014–2024 (Tables 4 and 5). We evaluated the methods against the 249 growth rates observed in 2025. Of the 423 non-market training assets, 416 had complete holdout histories; GARCH results covered 386 because its thirty non-stationary training fits remained excluded (Table 4). For each replication, we generated a 249-day SPY path from the training-period SPY marginal and used the same simulated market path in all asset compositions. In this evaluation, the correction used the observed training-period market variance in place of the simulated path variance σ_m^2 in Eqs. (8)–(12); the generator variance was still computed from each asset draw. The residual scale therefore budgeted for the training market variance, and variation in the simulated market variance could move the composed variance away from the generator target even when the sample cross-covariance was zero. Observed 2025 SPY and asset returns were used only for scoring. We repeated the KS, AD, Wasserstein-1, variance, kurtosis, and one-day left-tail Value-at-Risk (VaR) evaluations with 100 paths per ticker. For each VaR threshold, we also used Kupiec's unconditional-coverage test to compare the observed number of exceedances with the nominal rate (Kupiec, 1995). Because goodness-of-fit tests have less power on 249 observations than on 2,766, we also compared each synthetic path with a randomly selected contiguous 249-day block from the training history. We first averaged replication-level results within ticker and then summarized across tickers so each asset received equal weight. No 2025 data were used for fitting. The 249-day training blocks provided a comparison at the same sample length as the holdout period.

Jump-enabled multi-asset comparison.. To test whether the temporal benefit of the single-asset jump mechanism survived composition, we compared three settings: jumps disabled, jumps enabled only in the SPY market generator, and jumps enabled in both SPY and every asset generator (Table 6). We froze the existing closing-price state models and SIM calibrations. Enabled models received the SPY settings of Table 1, $(\epsilon, \lambda) = (10^{-4}, 90)$, $p_{\text{neg}} = 0.52$, and $N_{\text{tail}} = 5$, without per-asset tuning. The SPY calibration used volume-weighted average prices, so its transfer also changed the price convention. We simulated the market in both the 2,766-day training comparison and the 249-day holdout comparison; the preceding six-method training comparison instead used observed SPY. In both windows of the jump comparison, the correction used each simulated market path's variance and each asset draw's variance. Thus, the jump comparison also differed from the six-method holdout protocol, which kept the market variance fixed at its training value. Within each experiment, the variance convention was the same across all compared jump settings. This comparison assessed whether the SPY settings improved the composed paths without recalibrating the individual asset models. For each jump setting, naive and corrected composition received the same asset and market draws (Table 6). One market path was shared across all assets in each replication. We generated 250 paths for each of four seeds (1234, 2345, 3456, and 4567), giving 1,000 replications per asset, setting, and method. We measured absolute-return ACF-MAE over lags 1–25, with lags 1–60 as a secondary check, and repeated the marginal and variance diagnostics. These common lag windows differ from the single-asset comparison's 252-lag score. We included all paths in the primary summaries and estimated Monte Carlo uncertainty from replication-level cross-ticker averages, keeping assets together under their shared market path. The intervals describe simulation uncertainty conditional on the fitted models and observed histories, not uncertainty across market regimes.

4. Results

4.1. Single-asset generator validation

We first evaluated whether the asset generator captured the return distribution and volatility clustering that motivated its use in composition (Figs. 1 and 2; Supplementary Table S1). The observed SPY series had heavy tails and persistent absolute-return autocorrelation in both the training and holdout windows. Individual raw-return correlations were small, although the Ljung–Box test detected joint serial dependence (Supplementary Table S1). Calibration against the 2014–2024 history selected jump probability and mean duration $(\epsilon, \lambda) = (10^{-4}, 90)$, with similar objective values at nearby grid points (Eq. (3); Table 1; Algorithm 1). Ordinary states lasted only 1–2 steps, and the counted transition matrix implied absolute-return autocorrelation that decayed to near zero within a week (Supplementary Fig. S3). The selected episodes extended visits to extreme states, providing a mechanism for more persistent volatility. To assess whether the generator comparison depended on the state partition, we varied the number of states (Supplementary Table S7). The primary SPY analysis used $N = 100$ states, while sweeps from $N = 30$ to 200 changed HMM-NJ pass rates by less than one percentage point (Supplementary Table S7). The HMM-WJ model showed a tradeoff as the partition changed: narrower states improved autocorrelation accuracy but lowered the AD pass rate. At $N = 350$, some states were never visited and their emission parameters could not be estimated (Online Appendix S7). The selected $N = 100$ lay within the range that supported well-populated states for the 2,766-day history. That choice did not establish an appropriate resolution for shorter histories, which would provide fewer observations per state. The sensitivity checks therefore supported the fitted generator at the observed sample length without establishing one state count for every asset history.

We compared the calibrated HMM-WJ with seven alternative generators using 1,000 paths of 2,766 trading days (Table 2; Fig. 3). The HMM-NJ and HMM-WJ models had KS pass rates of 99.3% and 98.3%, respectively, giving the strongest overall distributional fit among the fitted parametric and latent-state generators. Bootstrap resampling passed every distribution test but did not reproduce volatility clustering, while Gaussian sampling passed none. The GARCH(1,1) model had the lowest mean absolute error in the autocorrelation function (ACF-MAE), 0.031, but rarely passed the distribution tests. The gated recurrent unit (GRU) (Cho et al., 2014) showed a similar tradeoff. The HSMM-style baseline adapted from Bulla and Bulla (2006) reached an 82% KS pass rate but underestimated kurtosis by more than one third. Within the HMM, Student-t emissions produced kurtosis of 7.5, close to the observed 7.7, whereas Normal emissions underestimated it by about 29% (Supplementary Table S4). Both HMM variants also reproduced the observed skewness of -0.75 through

Table 2: **In-sample and out-of-sample model comparison for SPY.** Results summarize 1,000 simulated paths. Values in parentheses are standard errors, and bold entries are best among the fitted generative models within each panel; the empirical bootstrap is excluded from this highlighting. Metric definitions and standard-error calculations are given in Online Appendix S3.

Model	**KS (%)**	**AD (%)**	$\boldsymbol{\kappa}$	**ACF-MAE**	**Coverage (%)**	$\boldsymbol{W_1}$	**Hellinger**
In sample: 2014–2024 *(*2,766 *days; observed* $\kappa = 7.715$*)*							
Bootstrap	100.0 (<0.1)	99.7 (0.2)	7.6 (0.08)	0.060 (<0.001)	100.0 (<0.1)	0.062 (0.001)	0.042 (<0.001)
Gaussian	0.0 (<0.1)	0.0 (<0.1)	−0.0 (<0.01)	0.060 (<0.001)	13.1 (0.1)	0.399 (0.001)	0.148 (<0.001)
Laplace	44.0 (1.6)	43.3 (1.6)	3.0 (0.02)	0.060 (<0.001)	37.4 (1.5)	0.138 (0.001)	**0.072** (<0.001)
GARCH(1,1)	5.5 (0.7)	1.9 (0.4)	8.2 (0.37)	**0.031** (0.001)	29.3 (1.4)	0.304 (0.006)	0.100 (0.001)
GRU	0.6 (0.2)	0.2 (0.1)	5.7 (0.16)	0.036 (<0.001)	17.2 (0.5)	0.421 (0.003)	0.134 (<0.001)
HSMM	82.0 (1.2)	42.5 (1.6)	4.8 (0.08)	0.059 (<0.001)	68.7 (1.1)	0.176 (0.001)	0.113 (<0.001)
HMM-NJ	**99.3** (0.3)	**98.2** (0.4)	8.1 (0.14)	0.059 (<0.001)	**100.0** (<0.1)	**0.081** (0.001)	**0.072** (<0.001)
HMM-WJ	98.3 (0.4)	94.8 (0.7)	**7.5** (0.09)	0.053 (<0.001)	**100.0** (<0.1)	0.097 (0.001)	0.074 (<0.001)
Out of sample: 2025 *(*249 *days; observed* $\kappa = 6.867$*)*							
Bootstrap	97.4 (0.5)	98.5 (0.4)	6.0 (0.18)	0.043 (<0.001)	**100.0** (<0.1)	0.232 (0.002)	0.205 (0.001)
Gaussian	62.2 (1.5)	46.3 (1.6)	−0.0 (<0.01)	0.043 (<0.001)	36.4 (1.9)	0.452 (0.002)	0.235 (0.001)
Laplace	88.0 (1.0)	92.9 (0.8)	2.7 (0.05)	0.043 (<0.001)	74.7 (1.0)	0.263 (0.002)	0.211 (0.001)
GARCH(1,1)	80.3 (1.3)	72.0 (1.4)	1.6 (0.07)	**0.026** (<0.001)	96.0 (1.3)	0.507 (0.018)	0.232 (0.001)
GRU	71.8 (1.4)	44.8 (1.6)	2.9 (0.10)	0.033 (<0.001)	77.8 (2.9)	0.488 (0.005)	0.240 (0.001)
HSMM	96.2 (0.6)	**96.7** (0.6)	4.1 (0.13)	0.042 (<0.001)	**100.0** (0.2)	0.287 (0.002)	0.239 (0.001)
HMM-NJ	**96.6** (0.6)	96.1 (0.6)	**6.2** (0.15)	0.041 (<0.001)	**100.0** (<0.1)	**0.259** (0.003)	**0.207** (0.001)
HMM-WJ	95.4 (0.7)	95.3 (0.7)	**6.2** (0.15)	0.040 (<0.001)	**100.0** (<0.1)	0.275 (0.005)	0.210 (0.001)

Model abbreviations are generalized autoregressive conditional heteroskedasticity (GARCH), gated recurrent unit (GRU), hidden semi-Markov model (HSMM), hidden Markov model with no jumps (HMM-NJ), and hidden Markov model with jumps (HMM-WJ). Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) report pass rates at $\alpha = 0.05$; coverage is the fraction of empirical quantiles inside the synthetic 90% envelope. κ is simulated excess kurtosis. Mean absolute error in the autocorrelation function (ACF-MAE) is computed for $|G_t|$ through lag 252. Larger values are preferred for KS, AD, and coverage; smaller values are preferred for ACF-MAE, W_1, and Hellinger distance; κ is compared with the observed value.

unequal occupancy of negative and positive states (Supplementary Table S1). The temporal comparison showed the cost of retaining that marginal fit. Jump episodes improved temporal fit, with a small reduction in distributional pass rates (Table 2; Fig. 3b). The HMM-WJ model reduced ACF-MAE from 0.059 for HMM-NJ to 0.053. About 24% of paths contained an episode (Supplementary Fig. S2); paths without an episode were equivalent to HMM-NJ. The episode share, the fraction of forced steps, and the jump-path autocorrelation agreed with the calculations in Online Appendix S4 (Supplementary Tables S2 and S3). To examine the tradeoff, we resampled the fitted ensemble with the fraction of jump-active paths ranging from zero to one. Increasing that fraction lowered autocorrelation error but also lowered KS and AD pass rates. Kurtosis declined through the observed value, while the Hill upper-tail index remained stable. The fitted ensemble retained approximately one quarter jump-active paths, balancing kurtosis and autocorrelation error. To test whether the distributional fit extended beyond calibration, we compared frozen-model paths with the 249 held-out SPY returns from 2025. The HMM-NJ and HMM-WJ models retained KS pass rates of 96.6% and 95.4%, with kurtosis close to the observed value (Table 2; Supplementary Fig. S2). Separate calibrations for NVDA, JNJ, and JPM also showed strong training fit but more variable holdout performance

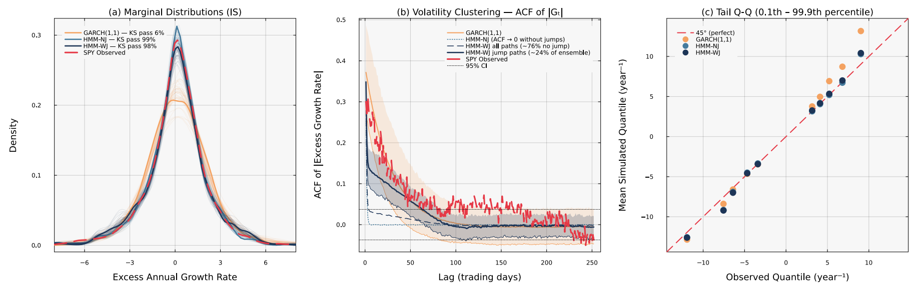

Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample Kolmogorov–Smirnov (KS) pass rates annotated per generator. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c):* tail quantile-quantile plot at the 0.1st–99.9th percentile region per generator.

(Supplementary Table S10). These findings supported the HMM as a source of heavy-tailed asset paths, while the benefit of the jump mechanism depended on the metric and evaluation period. We used the no-jump variant for the primary composition comparison and evaluated the transfer of jump settings separately in the multi-asset jump comparison.

4.2. Variance-corrected multi-asset composition

We compared six composition methods across 423 non-market assets to test whether the correction preserved generator variance and calibrated market loading (Table 3). Naive composition inflated variance by adding the market term to a draw that already represented total asset variance. The inflation increased with market exposure: at high β, composed variance exceeded generator variance by roughly 30–60%, and the naive KS pass rate fell toward zero above $\beta \approx 0.7$. The corrected method kept the variance ratio near one and retained substantially higher pass rates across the β range (Fig. 4). The cross-covariance omitted from the variance calculation was small: its normalized universe mean was 1.53×10^{-4}, with per-ticker values between -0.0061 and 0.0061 (Supplementary Fig. S1). The diagnostic supported the approximation used to derive the scale correction. The correction improved distributional fit while recovering the market loading with accuracy similar to the other methods (Table 3). All six methods had median absolute β errors between 0.017 and 0.022 (Table 3). Centering also kept the median absolute intercept error at $0.002\,\mathrm{yr}^{-1}$ for the naive and corrected methods; residual-fit methods retained finite-path variation in their residual means.

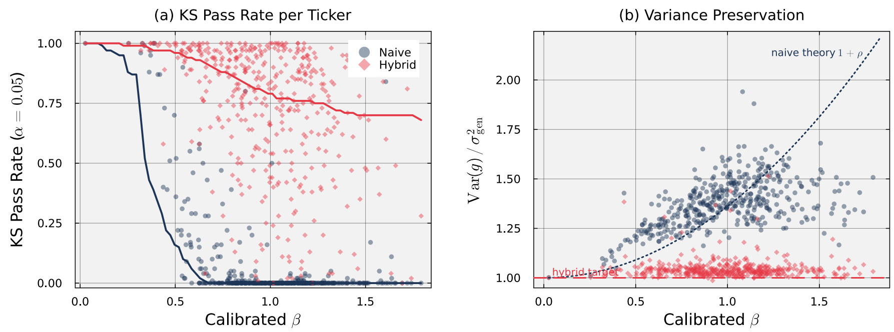

Figure 4: **Effect of the hybrid variance correction on each ticker's Kolmogorov–Smirnov pass rate and variance ratio.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Panel (a):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Panel (b):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ for the same two methods. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen} and the SPY annualized variance $\Delta t\, \sigma^2_m = 0.030\ \mathrm{yr}^{-1}$ of the closing-price series; the dashed line marks a variance ratio of one.

Table 3: **Aggregate scorecard across 423 non-market tickers and 100 replications per ticker.** Errors, distances, and tail metrics are medians per method. Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass fractions are pooled across replications. $|\hat{\alpha} - \alpha|$, $|\hat{\beta} - \beta|$, and $|\hat{R}^2 - R^2_{\mathrm{real}}|$: absolute deviation between ordinary least-squares (OLS) recovery and calibration (intercept error has units of yr^{-1}). KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. SIM denotes the single-index model, and GARCH denotes generalized autoregressive conditional heteroskedasticity.

Method	$\lvert\hat{\alpha} - \alpha\rvert$	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\mathrm{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.002	0.022	0.087	7.1	3.5	0.700	7.767	0.369
Gaussian SIM	0.048	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.002	0.018	0.021	73.6	64.6	0.249	7.165	0.365
JumpHMM-on-residuals	0.049	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.042	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.047	0.017	0.036	67.4	61.4	0.267	6.069	0.317

The methods differed more in distributional fit than in factor recovery (Table 3; Supplementary Fig. S6). Gaussian SIM had the lowest median R^2 error, but almost never passed the distribution tests and had excess kurtosis an order of magnitude below the observed value. The corrected method passed 73.6% of KS comparisons and retained heavy tails, with median excess kurtosis of 7.2 and a Hill index of 0.37 (Supplementary Fig. S6). Its median R^2 error fell between those of the naive and Gaussian methods because the main correction targeted generator variance rather than observed R^2. Preserving variance therefore improved the marginal fit without making every fitted property an exact target. Fitting directly to regression residuals provided a competitive alternative because those residuals already excluded the market component (Table 3). The three residual-fit methods had KS pass rates ranging from 67.4% for GARCH(1,1)-t to 75.8% for JumpHMM-on-residuals (Table 3). The corrected rate was about two percentage points below JumpHMM-on-residuals and slightly above block bootstrap. JumpHMM-on-residuals and block bootstrap had slightly lower Wasserstein distances, while JumpHMM-on-residuals matched the corrected method's kurtosis and Hill index almost exactly. The block-bootstrap and GARCH residual fits produced lighter tails. The corrected method thus approached the best residual-fit method without requiring another generator fit for each asset. The training comparison established the value of reuse relative to naive composition, but held-out histories were needed to assess whether that improvement persisted beyond the fitting period.

4.3. Holdout performance and VaR coverage

To test out-of-sample generalization without refitting or look-ahead bias, we froze the fitted models and SIM parameters before evaluating them against 416 complete asset histories from 2025 (Table 4). The corrected method reached an 87.0% mean cross-ticker KS pass rate, against 83.5% for centered naive composition, and its median composed-to-observed variance ratio was 0.97, whereas naive composition remained inflated at 1.29. JumpHMM-on-residuals reached an 86.2% KS pass rate and a slightly lower median W_1 than the corrected method. Matched-length training blocks produced lower pass rates for every method. The higher 2025 pass rates did not arise solely from comparing samples of 249 rather than 2,766 observations. The average pass rates concealed substantial differences among tickers (Fig. 5). Most ticker-level rates were near the upper end of the range, but every method retained a lower tail of weak fits. For the corrected method, many tickers improved relative to their matched-length training blocks, while a subset deteriorated out of sample. The held-out marginal improvement was broad but did not ensure reliable extreme-loss thresholds. To test whether marginal fit translated into reliable tail thresholds, we used the frozen synthetic paths to estimate per-ticker one-day left-tail Value-at-Risk (VaR) thresholds and counted exceedances on the real 2025 histories (Table 5). At $\alpha = 0.95$, the corrected and JumpHMM-on-residuals methods produced mean exceedance rates of 5.97% and 6.07%, respectively, against the 5% nominal rate. At $\alpha = 0.99$, both rates were 1.67%, against the 1% nominal rate. Naive composition was closest at the 99% level (1.21%), while Gaussian SIM had the largest exceedance rate (2.03%). The short holdout contained only about 2–3 expected 99% exceedances per ticker, so we interpreted the corresponding Kupiec pass rates and cross-ticker dispersion as low-power checks rather than precise rankings. The corrected method improved marginal fit over naive composition, but its tail thresholds still produced more exceedances than the nominal rates. The coverage results therefore limited the risk interpretation of the distributional improvement.

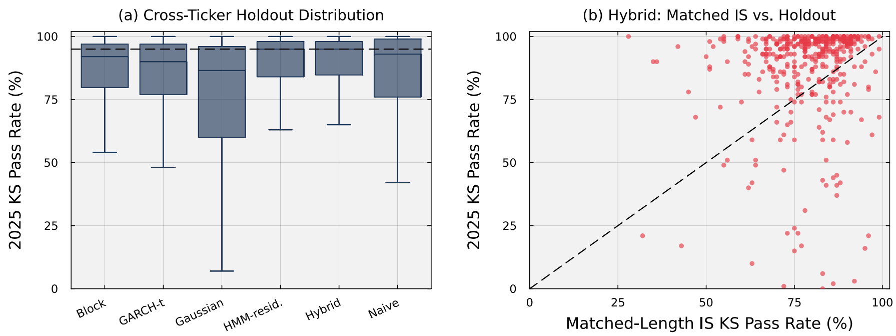

Figure 5: **Cross-ticker distribution of frozen-parameter out-of-sample marginal fit.** Panel (a) shows the distribution of per-ticker 2025 Kolmogorov–Smirnov (KS) pass rates across 100 replications; the dashed line marks 95%. Panel (b) compares each ticker's hybrid KS pass rate against a matched-length training block with its pass rate against the 2025 holdout. The diagonal denotes equal performance.

Table 4: **Frozen-parameter multi-asset evaluation on the** 2025 **holdout.** Results cover 416 non-market tickers and 100 replications per ticker; generalized autoregressive conditional heteroskedasticity (GARCH) covers 386 tickers. Pass rates are first averaged within ticker and then across tickers. The *matched in-sample (IS)* columns compare each synthetic path with a randomly selected contiguous 249-day block from the 2014–2024 training history, matching the holdout sample length. W_1 and the variance ratio are medians across ticker-level medians; the variance ratio is synthetic variance divided by observed 2025 variance. The hybrid correction used the training-period market variance and each simulated asset draw's variance.

	KS pass (%)		AD pass (%)			
Method	matched IS	2025	matched IS	2025	W_1 (2025)	variance ratio
Naive	69.4	83.5	54.8	71.6	0.877	1.29
Gaussian SIM	57.4	73.9	44.9	64.6	0.866	0.97
Hybrid	78.8	87.0	66.2	79.7	0.716	0.97
JumpHMM-on-residuals	77.9	86.2	65.6	80.1	0.708	0.98
Block bootstrap	76.3	84.0	64.0	76.5	0.715	0.90
GARCH(1,1)-t	74.5	82.6	62.6	75.0	0.733	0.91

Table 5: **Out-of-sample per-ticker one-day left-tail Value-at-Risk coverage across the six composition methods.** VaR thresholds at coverage level α were estimated from frozen-parameter synthetic paths and exceedances were counted against the real 2025 growth-rate history. The hybrid correction used the same training-period market variance convention as Table 4. *rate*: mean of the per-ticker mean exceedance rates (nominal: $1-\alpha$). Cross-ticker standard deviation (SD) is reported in percentage points. *Kupiec pass*: mean within-ticker fraction of replications whose unconditional-coverage test p-value exceeds 0.05.

	$\alpha = 0.95$			$\alpha = 0.99$		
Method	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	4.37	2.15	65.8	1.21	0.76	78.9
Gaussian SIM	4.95	2.26	70.0	2.03	1.17	72.1
Hybrid	5.97	2.43	66.3	1.67	0.93	73.4
JumpHMM-on-residuals	6.07	2.59	67.0	1.67	0.95	74.5
Block bootstrap	6.13	2.56	63.7	1.84	1.01	73.4
GARCH(1,1)-t	5.79	2.47	62.4	1.80	1.02	72.0

4.4. Jump-enabled multi-asset composition

To test whether jump episodes improved the composed paths, we compared jumps disabled, market-only jumps, and market-plus-asset jumps under naive and corrected composition (Table 6). With corrected composition, market-only jumps reduced the training absolute-return ACF-MAE over lags 1–25 from 0.13569 to 0.12844, a 5.3% reduction. The KS pass rate changed from 73.94% to 73.64%, with a Monte Carlo interval for the difference that included zero (Supplementary Table S12). Enabling jumps in both the market and asset generators reduced the ACF-MAE further, to 0.11581, a 14.6% reduction from the no-jump setting, but lowered the KS pass rate to 61.68%. Across all settings and both windows, the mean ratio of corrected to generator variance differed from one by less than 0.05%. The correction remained effective, while asset episodes reduced training autocorrelation error at the cost of marginal fit. The training-period temporal improvement did not carry into the 2025 holdout (Table 6). With corrected composition, market-only jumps raised the 25-lag ACF-MAE from 0.07330 to 0.07525, a 2.7% increase, while market-plus-asset jumps raised it to 0.07832, a 6.9% increase (Table 6). The corresponding KS pass rates were 85.92%, 85.88%, and 84.43%. The Monte Carlo intervals excluded zero for both increases in ACF error, and the 60-lag results agreed (Supplementary Table S12). The jump-active market paths represented only 2.2% of the simulated one-year ensemble (Supplementary Table S13), so the holdout comparison tested the fitted mixture of ordinary and jump-active paths. The correction worked with jumps enabled, but the transferred SPY settings did not improve temporal fit in 2025. The short holdout and low episode frequency limited what the comparison could establish about behavior in other market periods.

Table 6: **Jump settings under naive and corrected multi-asset composition.** Each setting used 1,000 replications per asset across four seeds, with identical asset and market draws for the two composition methods. SPY was simulated in both windows, and the correction used each simulated market path's variance and each asset draw's variance. Entries are equal-weight cross-ticker means of per-path metrics. Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) report non-rejection rates at $\alpha = 0.05$. ACF_{25} and ACF_{60} are mean absolute errors in the absolute-return autocorrelation function through lags 25 and 60; lower values indicate better temporal fit. $V_{\text{gen}} = \mathrm{Var}(g_i)/\sigma^2_{\text{gen},i}$ compares composed variance with the active generator's variance, rather than observed variance.

Jump setting	Method	KS (%)	AD (%)	ACF_{25}	ACF_{60}	V_{gen}
Training: 2014–2024, 423 *assets,* 2,766 *days*						
Off	Naive	6.93	3.38	0.13585	0.09492	1.33273
Off	Corrected	73.94	68.54	0.13569	0.09485	1.00011
Market only	Naive	6.28	3.18	0.13197	0.09208	1.34938
Market only	Corrected	73.64	67.81	0.12844	0.08969	1.00032
Market + assets	Naive	5.25	2.56	0.11727	0.08264	1.33445
Market + assets	Corrected	61.68	53.33	0.11581	0.08154	1.00018
Holdout: 2025, 416 *assets,* 249 *days*						
Off	Naive	82.38	70.14	0.07322	0.06683	1.36081
Off	Corrected	85.92	78.76	0.07330	0.06687	1.00041
Market only	Naive	81.76	69.60	0.07347	0.06698	1.36638
Market only	Corrected	85.88	78.72	0.07525	0.06796	1.00045
Market + assets	Naive	80.27	68.19	0.07672	0.06885	1.36032
Market + assets	Corrected	84.43	77.21	0.07832	0.06973	1.00039

4.5. Sensitivity analyses

We examined the tracker and clipping rules under conditions where they could affect the result (Supplementary Figs. S4 and S5; Supplementary Table S5). In the observed universe, 421 assets used the variance-preserving branch and two, QQQ and SPYG, used the R^2-preserving branch; none required clipping (Supplementary Figs. S5 and S4; Supplementary Table S5). Both trackers had 100% KS pass rates, while the highest-R^2 individual stock, BLK, fell about 0.16 below the tracker threshold of 0.80. Because only two assets used that branch, we tested a synthetic-tracker grid with $(\beta_{\text{true}}, R^2_{\text{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$. Each cell had a 100% KS pass rate and recovered its target loading and R^2 (Supplementary Table S6). These checks supported the tracker rule across the tested grid, while its empirical evaluation remained limited to two assets. Greater market exposure made the composed distribution more dependent on the market path (Fig. 4b; Supplementary Table S8). Grouping assets by β quartile showed that corrected KS pass rates fell from 87.6% in the lowest quartile to 65.9% in the highest (Supplementary Table S8). To test the loading adjustment directly, we multiplied the market growth-rate path by $\gamma \in \{1, 2, 3\}$ while holding the calibrated marginals fixed (Supplementary Table S9). At $\gamma = 1$, every asset had $\rho_i < 1 - f = 0.90$, so clipping was unnecessary. At $\gamma = 2$ and 3, clipping applied to 76.4% and 95.2% of assets. The adjustment therefore activated as intended when the market contribution would otherwise consume too much of the target variance. At observed market variance, varying the residual floor f did not change branch assignments, and varying the tracker threshold changed only the two tracker assignments (Supplementary Table S5).

5. Discussion

The variance correction allowed fitted full-return generators to be reused in a single-index model without adding market variance twice (Fig. 4; Table 3). Its practical value was avoiding a separate generator fit for each asset's regression residuals while achieving comparable distributional fit. The corrected method passed 73.6% of the full-length training KS comparisons and retained heavy tails (Table 3). On the 2025 holdout, it raised the mean cross-ticker KS pass rate from 83.5% under centered naive composition to 87.0% and held the median ratio of synthetic to observed variance at 0.97 (Table 4). JumpHMM-on-residuals had a slightly higher pass rate in training and a slightly lower rate in the holdout. The comparison supported reuse as a practical option when full-return generators are already available; fitting directly to residuals remained a competitive alternative. Neither method was best for every asset, and the average improvement concealed a subset of corrected paths whose fit deteriorated relative to matched-length training blocks. The validation established closer marginal fit, but it did not establish reliable tail coverage (Table 4; Table 5). The corrected method and JumpHMM-on-residuals both had a one-day left-tail 99% VaR exceedance rate of 1.67%, above the nominal 1% rate (Table 5). The 249-day holdout contained too few expected extreme events per asset for a precise ranking. Distributional pass rates also require care because the two-sample KS and AD tests assume independent observations, whereas several evaluated generators produced serial dependence. We therefore interpreted pass rates alongside distributional distances, tail summaries, and autocorrelation errors. Matched-length training blocks reduced the difference in test power between the training and holdout comparisons, but the observed 2025 history still represented only one market period. The evidence supported improved variance and distributional fit across the evaluated universe, with substantial uncertainty about performance in other periods and for individual assets.

The jump comparison showed that improving the single-asset generator did not automatically improve the composed paths (Table 2; Table 6). On SPY, HMM-WJ reduced absolute-return autocorrelation error relative to HMM-NJ while retaining high distributional pass rates. Other generators achieved lower autocorrelation error or higher pass rates, so the single-asset result supported a tradeoff between temporal and marginal fit (Table 2). With multi-asset composition, market-only jumps improved training temporal fit with little change in the KS pass rate, whereas adding asset-level episodes produced a larger temporal improvement at a substantial marginal-fit cost. Both settings increased autocorrelation error in 2025, although the variance correction remained effective (Table 6). These tests transferred one SPY calibration from volume-weighted average prices to closing-price models. Separate calibrations for NVDA, JNJ, and JPM had strong in-sample fit, but three assets did not establish a general calibration rule (Supplementary Table S10). The asset episodes also came from full-return models and were not identified separately as idiosyncratic events. Per-asset tuning or distinct shared and asset-specific episode models could change the tradeoff, but neither was tested here.

The correction's effect on the return distribution depended on the asset's market exposure (Fig. 4; Supplementary Table S8). Corrected KS pass rates declined from 87.6% in the lowest β quartile to 65.9% in the highest because the market path accounted for more of the composed variance (Supplementary Table S8). Preserving total variance therefore did not preserve the complete shape of the original return distribution. The loading adjustment was unnecessary at observed market variance but activated for most assets when market volatility was doubled or tripled (Supplementary Table S9). Only two tracker assets used the separate R^2-preserving branch (Supplementary Table S5), so evidence for that branch relied mainly on the synthetic-tracker checks. These properties also depended on generating asset and market draws independently. If their covariance were nonzero, the variance calculation would need an additional cross-term. The correction thus addressed a specific composition problem, with explicit assumptions about the supplied paths and adjustments when the market term consumed too much of the target variance. Dependence among asset-specific residuals remained outside the composition rule (Algorithm 2; Online Appendix S2). A shared market path linked same-day asset returns and could produce simultaneous large moves, but independent residual draws omitted the sector and style dependence present in real assets. Omitting positive residual covariance can overstate the diversification return from periodic rebalancing (Booth and Fama, 1992; Bouchey et al., 2015; Markowitz, 1976); Online Appendix S2 derives the contribution for this construction. The limitation also applies to the residual-fit baselines when their residuals are generated independently. Shifting paths leaves cross-asset covariance unchanged, and rescaling independent residuals cannot create the missing dependence. A synchronized multivariate block bootstrap, residual factor model, or copula could address that omission, while separate univariate models would improve only within-asset behavior. The present evaluation measured per-asset fit and VaR coverage, so it did not validate portfolio-level risk or rebalancing performance. Those uses require a joint residual model and direct evaluation of the resulting dependence.

6. Conclusion

We developed a variance-corrected single-index method that reuses full-return generators without shifting the calibrated mean or counting market variance twice (Fig. 4; Table 3). Centering and rescaling allowed the generated asset paths to be combined with a shared market path without fitting a second generator to each asset's regression residuals. Across 423 non-market assets, the method recovered calibrated market loadings with low error and retained heavy-tailed return distributions. On 416 complete asset histories from 2025, it improved the mean KS pass rate over centered naive composition and brought the median ratio of synthetic to observed variance close to one (Table 4). Residual-fit methods remained competitive, and neither the corrected method nor the best residual-fit alternative achieved nominal 99% VaR coverage (Table 5). The results supported reuse of existing generators as a practical alternative to fitting separate residual models, with validation still needed for each intended application. The jump experiments established that the correction also worked when the generators included extended episodes in extreme-return states (Table 6). The SPY jump settings improved multi-asset temporal fit in training but worsened it in the 2025 holdout, showing that the single-asset benefit did not transfer automatically. Correcting variance also left dependence among asset-specific residuals unresolved. Independent residual draws omit relationships that matter for portfolio risk and can overstate the return from rebalancing when positive residual covariance is missing. The composition rule therefore provides one component of a multi-asset simulator: it controls how generator variance is combined with the market factor, while portfolio-level applications require a joint residual model and validation of both cross-asset dependence and temporal behavior.

Code and data availability

The experiment scripts, evaluation code (six-method comparison and left-tail Value-at-Risk coverage check), cached inputs and result summaries, and instructions for regenerating the fitted per-asset models are publicly available. The underlying model-fitting library is released separately and pinned as a dependency of the experiment code. Identifying repository links have been withheld from this manuscript for double-anonymized review and are provided to the editorial office on the separate title page.

Declaration of generative artificial intelligence use in the writing process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, LaTeX formatting, and the implementation and checking of experiment code. The authors reviewed and edited the assisted work and take full responsibility for the content of the publication.

References

Aas, K., Czado, C., Frigessi, A., Bakken, H., 2009. Pair-copula constructions of multiple dependence. Insurance: Mathematics and Economics 44, 182–198.

Assefa, S.A., Dervovic, D., Mahfouz, M., Tillman, R.E., Reddy, P., Veloso, M., 2020. Generating synthetic data in finance: Opportunities, challenges and pitfalls, in: Proceedings of the First ACM International Conference on AI in Finance, pp. 1–8. doi:10.1145/3383455.3422554.

Bauwens, L., Laurent, S., Rombouts, J.V.K., 2006. Multivariate GARCH models: A survey. Journal of Applied Econometrics 21, 79–109. doi:10.1002/jae.842.

Bilmes, J.A., 1998. A Gentle Tutorial of the EM Algorithm and Its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models. Technical Report TR-97-021. International Computer Science Institute.

Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 31, 307–327. doi:10.1016/0304-4076(86)90063-1.

Booth, D.G., Fama, E.F., 1992. Diversification returns and asset contributions. Financial Analysts Journal 48, 26–32. doi:10.2469/faj.v48.n3.26.

Bouchey, P., Nemtchinov, V., Wong, T.K.L., 2015. Volatility harvesting in theory and practice. Journal of Wealth Management 18, 89–100. doi:10.3905/jwm.2015.18.3.089.

Bulla, J., Bulla, I., 2006. Stylized facts of financial time series and hidden semi-Markov models. Computational Statistics & Data Analysis 51, 2192–2209. doi:10.1016/j.csda.2006.07.021.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics. pp. 1724–1734. doi:10.3115/v1/D14-1179.

Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statistical issues. Quantitative Finance 1, 223–236. doi:10.1080/713665670.

Dias, J.G., Vermunt, J.K., Ramos, S., 2010. Mixture hidden Markov models in finance research, in: Advances in Data Analysis, Data Handling and Business Intelligence, Springer. pp. 451–459. doi:10.1007/978-3-642-01044-6_41.

Embrechts, P., McNeil, A.J., Straumann, D., 2002. Correlation and dependence in risk management: Properties and pitfalls, in: Risk Management: Value at Risk and Beyond. Cambridge University Press, pp. 176–223. doi:10.1017/CBO9780511615337.008.

Engle, R., 2002. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. Journal of Business & Economic Statistics 20, 339–350. doi:10.1198/073500102288618487.

Engle, R.F., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica 50, 987–1007.

Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33, 3–56. doi:10.1016/0304-405X(93)90023-5.

Glasserman, P., 2003. Monte Carlo Methods in Financial Engineering. Springer. doi:10.1007/978-0-387-21617-1.

Grinstead, C.M., Snell, J.L., 1997. Introduction to Probability. American Mathematical Society.

Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica 57, 357–384. doi:10.2307/1912559.

Hamilton, J.D., 2018. Regime switching models. The New Palgrave Dictionary of Economics , 11421–11426doi:10.1057/978-1-349-95189-5_2459.

Hamilton, J.D., Susmel, R., 1994. Autoregressive conditional heteroskedasticity and changes in regime. Journal of Econometrics 64, 307–333. doi:10.1016/0304-4076(94)90067-1.

Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies 6, 327–343. doi:10.1093/rfs/6.2.327.

Jordon, J., Szpruch, L., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S.N., Weller, A., 2022. Synthetic data: What, why and how? arXiv preprint arXiv:2205.03257 .

Kim, S., Shephard, N., Chib, S., 1998. Stochastic volatility: Likelihood inference and comparison with ARCH models. The Review of Economic Studies 65, 361–393.

Klaassen, F., 2002. Improving GARCH volatility forecasts with regime-switching GARCH. Empirical Economics 27, 363–394. doi:10.1007/s001810100100.

Kotz, S., Kozubowski, T.J., Podgórski, K., 2001. The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance. Birkhauser.

Kou, S.G., 2002. A jump-diffusion model for option pricing. Management Science 48, 1086–1101. doi:10.1287/mnsc.48.8.1086.166.

Kupiec, P.H., 1995. Techniques for verifying the accuracy of risk measurement models. The Journal of Derivatives 3, 73–84. doi:10.3905/jod.1995.407942.

Kwon, S., Lee, Y., 2024. Can GANs learn the stylized facts of financial time series?, in: Proceedings of the 5th ACM International Conference on AI in Finance, pp. 126–133. doi:10.1145/3677052.3698661.

Mandelbrot, B., 1963. The variation of certain speculative prices. The Journal of Business 36, 394–419.

Markowitz, H.M., 1976. Investment for the long run: New evidence for an old rule. Journal of Finance 31, 1273–1286. doi:10.1111/j.1540-6261.1976.tb03213.x.

Merton, R.C., 1976. Option pricing when underlying stock returns are discontinuous. Journal of Financial Economics 3, 125–144. doi:10.1016/0304-405X(76)90022-2.

Ni, H., Szpruch, L., Sabate-Vidales, M., Xiao, B., Wiese, M., Liao, S., 2021. Sig-wasserstein GANs for time series generation, in: Proceedings of the Second ACM International Conference on AI in Finance, ACM. pp. 1–8. doi:10.1145/3490354.3494393.

Patton, A.J., 2006. Modelling asymmetric exchange rate dependence. International Economic Review 47, 527–556. doi:10.1111/j.1468-2354.2006.00387.x.

Politis, D.N., Romano, J.P., 1994. The stationary bootstrap. Journal of the American Statistical Association 89, 1303–1313. doi:10.1080/01621459.1994.10476870.

Rabiner, L., Juang, B., 1986. An introduction to hidden Markov models. IEEE ASSP Magazine 3, 4–16. doi:10.1109/MASSP.1986.1165342.

Rizzato, M., Wallart, J., Geissler, C., Morizet, N., Boumlaik, N., 2023. Generative adversarial networks applied to synthetic financial scenarios generation. Physica A: Statistical Mechanics and its Applications 623, 128899. doi:10.1016/j.physa.2023.128899.

Ross, S.A., 1976. The arbitrage theory of capital asset pricing. Journal of Economic Theory 13, 341–360. doi:10.1016/0022-0531(76)90046-6.

Rydén, T., Teräsvirta, T., Åsbrink, S., 1998. Stylized facts of daily return series and the hidden Markov model. Journal of Applied Econometrics 13, 217–244. doi:10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V.

Sharpe, W.F., 1963. A simplified model for portfolio analysis. Management Science 9, 277–293. doi:10.1287/mnsc.9.2.277.

Stein, E.M., Stein, J.C., 1991. Stock price distributions with stochastic volatility: An analytic approach. The Review of Financial Studies 4, 727–752.

Stenger, M., Bauer, A., Prantl, T., Leppich, R., Hudson, N., Chard, K., Foster, I., Kounev, S., 2024. Thinking in categories: A survey on assessing the quality for time series synthesis. Journal of Data and Information Quality 16, 1–32. doi:10.1145/3666006.

Takahashi, S., Chen, Y., Tanaka-Ishii, K., 2019. Modeling financial time-series with generative adversarial networks. Physica A: Statistical Mechanics and Its Applications 527, 121261. doi:10.1016/j.physa.2019.121261.

Toth, D., Jones, B., 2019. Against the norm: Modeling daily stock returns with the Laplace distribution. arXiv preprint arXiv:1906.10325 .

Wiese, M., Knobloch, R., Korn, R., Kretschmer, P., 2020. Quant GANs: Deep generation of financial time series. Quantitative Finance 20, 1419–1440. doi:10.1080/14697688.2020.1730426.

Yoon, J., Jarrett, D., van der Schaar, M., 2019. Time-series generative adversarial networks, in: Advances in Neural Information Processing Systems, pp. 5509–5519.